Accelerating Diagnosis Through NLP-Driven Text Classification of Medical Notes

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1 Introduction

Electronic health records (EHR) store information on patient status, labs, medication, procedures, and examinations. Data about electronic health records are generally well-documented and thoroughly illustrate patient symptoms and medicated vitals [4]. The transition from physical medical records to EHR has been increasing in the United States as it promotes a healthcare system that is connected and well-informed about the patient.

With this change, there are also efforts to improve the data and consistency in EHR. EHR data is highly complex with the breadth and number of patients. Determining the most accurate or best diagnosis can be difficult with the structure of clinical information. The vast amounts of EHR data also present the opportunity for machine learning. EHR can be leveraged to improve predictions in patients health, specifically diagnosis. This project explores the capabilities of natural language processing and machine learning tools in predicting diagnosis using EHR.

1.1 Project Objectives

In this project, instead of focusing on a delivering a perfect model, we will direct our focus on exploring different features to improve predictions in patients' health. We first aim to focus on feature identification by identifying critical features in medical notes that help predict the top three most common diagnosis from the dataset, Pneumonia, Sepsis, and Congestive Heart Failure. To do this, we hope to experiment with different Natural Language Processing (NLP) techniques to explore varying analysis methods. From here we hope to expand our text classification into a usable Recurrent Neural Network (RNN) Model. Overall, we want to develop a system where our model continuously improves in performance as it trains on more data identified using NLP. We strive to build a model that can address the complexity of medical records, but above all provide the best solutions for improving patient health efficiently and effectively.

2 Background

As the quality of EHR and the development of **artificial intelligence (AI)** improve, there is a growing opportunity to apply both tools in creating a more advanced healthcare system. The implementation of AI in hospitals aims to increase efficiency, specificity, and quality of care for patients [7]. AI applications in healthcare enable predictive disease modeling, image analysis, and tailored patient care. Although AI is relatively novel in this setting, has the potential to become a powerful tool that can accelerate care delivery and improve outcomes.

Machine learning, a core component of AI, plays a crucial role in transforming healthcare data into actionable insights. In healthcare, **machine learning algorithms** can be trained on large datasets of patient records, enabling the identification of patterns that might be too complex for regular computer models or human analysts to detect.

For instance, AI models can predict the incidence of Diabetes Mellitus using routine health records. This model predicted up to 95% accuracy [5]. As noted, this example highlights how the vast data in EHRs, collected daily by hospitals, can be leveraged by AI to produce valuable insights for decision making.

While EHR proves to be invaluable for clinical care and research, it also provides significant challenges in real-world applications. One of the primary issues is with the collection of data and quality. Data can be collected from an unequal range of demographics and systemic bias can alter the treatment for patients. Marginalized subgroups may receive worse treatment and diagnosis results [1]. Systemic biases through the healthcare data can worsen existing inequalities and misidentify true healthcare needs. This leads to another issue: the implementation of EHR systems is inevitably vulnerable to errors. Several problems can occur with the usability of the system, the metadata, or formatting. In a case study, a gynecologist misread a positive mutation finding, in his patient, for a negative finding. "Eighteen months later, the patient was diagnosed with Stage III ovarian cancer" [3]. The doctor claimed to have had a lack of familiarity with the newly revised EHR system. This example showcases how a minor oversight of information resulted in a diagnostic error and severe outcome.

This case study involving EHR is just one instance of a larger problem: diagnostic errors. Diagnostic errors include medical diagnoses that are missed, incorrect, or delayed. Each year in the US, 100,000 patients either die or are permanently disabled as a result of diagnostic errors [6]. These are considered some of the most common, costly, and catastrophic medical mistakes. According to Johns Hopkins Armstrong Institute for Patient Safety and Quality, the top three major disease categories impacted are vascular events, infections, and cancers. Fifteen conditions within these top three categories alone account for nearly half of all the serious misdiagnosis harms [6]. The necessity to tackle and reduce diagnostic errors, coupled with the increasing about of EHR, and the evolving role of machine learning highlight an opportunity where data science can have great impact.

Natural Language Processing, a subfield of artificial intelligence that enables computers to understand and evaluate human language, can play a key role in enhancing EHR utility. NLP enables computers to process and analyze human language allowing the extraction of valuable insights from large, dense, and sometimes unstructured text data in EHRs. Machine learning is essential in NLP for training models to accurately interpret language for specific contexts in healthcare. By using machine learning, NLP models can efficiently classify text, detect potential diagnoses, and summarize patient histories - all to assist healthcare providers in making more well-informed decisions.

With continued training, machine learning models coupled with NLP, can improve over time, becoming more effective at identifying relevant patterns in EHR notes that can potentially be used to predict patient diagnosis, treatment, and medication. This project combines the potential of EHR data and NLP to produce a model that can help streamline the process of medical diagnosis. This is one approach to address the alarming number of diagnostic errors that occur annually. The lasting impacts should not be taken lightly, and require intentional and innovative solutions. In addition to the larger problem at hand, a NLP-driven machine learning model can help hospitals save time and resources and provide patients with the quality care they deserve while minimizing biases.

3 Methods

3.1 Data

This project uses the MIMIC-III dataset. Medical Information Mart for Intensive Care or MIMIC is a public database that contains de-identified health-related patient data for those admitted to the critical care units at Beth Israel Deaconess Medical Center. The data from MIMIC-III contains patient records from 2001-2012. For this project, we are interested in the NOTEEVENTS and ADMISSIONS tables. The NOTEEVENTS table contains information on patient ID, stay, time data for each medical note, category, description of note type, and note text data. This project will focus on the use of note text data found in the 'TEXT' column of the NOTEEVENTS data table. Feature engineering will be performed on the text data with natural language processing. The ADMISSIONS table contains information on the patient's admission to the hospital including demographics, location, time, and diagnosis. This project will focus on the 'DIAGNOSIS' column to perform categorical classification of patient notes.

The two tables can be joined on HADM_ID which is a unique code for each patient stay. Once the data is merged and includes the 'TEXT' and 'DIAGNOSIS' columns, the additional columns are removed except for ROW_ID and SUBJECT_ID. These columns act as identifiers and will not be used in the model training. The resulting table has the patient note text matched with the corresponding diagnosis label. Using these two columns, the model aims to predict diagnosis based on the language used in patient notes for various admission events.

The raw NOTEEVENTS data contains over 2 million rows. In order to reduce processing size, the data will be filtered to only include the top three diagnosis labels: Pneumonia, Sepsis, and Congestive Heart Failure. The data was randomly filtered to include balanced numbers for each label in order to prevent skewed results. As a result of the data processing, the data has about 54,000 rows.

For further exploration of model performance we prepared a data set with less common conditions. we filtered the data to contain three of the less common diagnoses in the data set. This was determined by identifying the top three conditions with under 1000 instances in the dataset. We chose 1000 as a standard threshold because it is significantly less than the top three diagnoses counts, but still large enough to train a model with. The selected conditions were stroke, renal failure, and wound infection. The data set has about 3,000 rows.

3.2 Machine Learning Approaches

Thus far, our team has implemented Logistic Regression and Bernoulli as a Naive Bayes classifier for our two Bag of Words models. We are using Logistic Regression as a primary model for it's effectiveness in binary classification. Logistic Regression is a linear model that optimizes coefficients for each feature by minimizing the error between predicted and actual labels. It is particularly useful for text classification, as it can handle high-dimensional feature spaces.

We are also testing with Naive Bayes, specifically BernoulliNB, because in text-based classification tasks, it assumes feature independence and is efficient computationally as well. Assuming feature independence treats the presence or absence of each word as independent of others. This is useful for text data as each word can independently contribute valuable information for the classification. Bernoulli Naive Bayes estimates the probability that a text belongs to each class based on the frequency of word presence across classes, making it efficient with binary Bag of Words features.

We also implemented TF-IDF (Term Frequency-Inverse Document Frequency) feature engineering for our models to statistically measure the importance of words and phrases in our text. TF-IDF reduces the weight of frequently occurring, but less informative words, like common stop words and amplifies the importance of rarer, more distinguishing words. In the note data we want to highlight medical terminology as opposed to common stop and filler words that will appear in all medical notes eg. "patient" or "pain." This enables the models to focus on terms that are likely more relevant to the three diagnoses of interest. A max_df of 0.7 was added to hyper-tune the filtering of less frequent/informative words as well as a min_df of 0.5 to remove more rare terms.

We first implemented a simple Logistic Regression and Naive Bayes Classification with the top three diagnoses as an initial proof of concept and benchmark. As we progressed into our project, we also looked at less frequently diagnosed diseases to see how our models performed on a small subset of data.

We utilized a dummy classifier as a baseline model to help gauge the true effectiveness of the fine-tuned models we implement. The dummy classifier established a minimum baseline without learning from the data. This will help truly assess our model's performance. If it is struggling to outperform the dummy classifier, then there might be issues including class imbalance. This will be especially important for our BoW implementation to determine if our features contain enough information for predictive modeling.

We defined a dictionary to assign severity weights of descriptive words such as "mild", "moderate", "severe", "critical", and "life-threatening" in order for the model to differentiate and prioritize certain symptoms to aid more precise diagnoses.

In addition, we defined a dictionary of synonymous medical terminologies so that they will be counted as the same term and therefore more accurately represent the weight of the symptom when the model makes predictions. We also built our own dictionary of custom stop words often found in medical notes, such as "patient", "reported", "history", and "observed", on top of the stop word hyper-parameter in the TF-IDF implementation.

Lastly, we experimented with recurrent neural networks (RNNs) to convert the sequential symptoms in a patient's notes to a pathway that predicts and detects symptoms of a disease more effectively. RNNs have the ability to have context about information they are dealing with because they can utilize information from previous positions in the sequence. We specifically used Long Short Term Memory network (LSTM) for its higher memory power advantage. Additionally, LSTMs are able to capture long-term dependencies which we realized would be especially helpful within the medical data notes to better track the diagnosis progression. Along with adjusting the layers we added dropout to minimize overfitting and tested out different batch sizes and class weight balancing.

3.3 Evaluation

From our current implementation we have produced a classification report that includes, accuracy, precision, recall and F1 score. This has given us a comprehensive view of the model's performance and offered us an overview of where our models need to improve. Overall, we conducted two main experiments:

- Bag of words classification exploration
- Recurrent neural network implementation

First, we built two Bag of Words Models on the (medical) notes in our dataset. One model looks at keywords at the sentence level of each note while the other looks at neighboring keywords as opposed to the entire note through N-Gram encoding. For each model, we split the data into the notes being the x-variable and the diagnosis being the y-variable. We also utilized the same training, validation, and test sets to build each model. Both Bag of Words models have stop words filtered out as apart of the algorithm. Additionally, for more accurate evaluations we want to try excluding more stop and filler words.

To further improve our Bag of Words model, we "injected knowledge" into the model by giving it "human rules" for it to perform better in a health context. This approach involved encoding domain-specific insights into the model, so it prioritizes and weighs words and phrases that are more likely to be diagnostically relevant. The "human rules" we implemented are:

- Synonyms and Medical Terminology: Medical language often includes synonyms and variations in terminology (e.g., "myocardial infarction" vs. "heart attack"). A rule can be designed to treat these terms as equivalent by assigning them the same weights or grouping them together.
- Symptom Severity: For symptoms that vary in severity, human rules can help determine the diagnostic weight assigned to terms like "mild," "moderate," or "severe." Severe symptoms might increase the likelihood of more serious conditions, guiding the model toward certain diagnoses.

Second, we evaluated several metrics for our Reccurent Neural Network. In addition to looking at the classification report and validation accuracy during each epoch of training, we observed the cross-entropy loss. Cross-entropy was best in classification tasks such as classifying the diagnoses in our data to measure how well the predicted probability distribution compared to the true probability distribution.

Success was primarily measured by comparing the performance of machine learning techniques from the Bag of Words and RNN models and looking at the f1-score as it provides a balanced evaluation of precision and recall. This was through metric computations where we calculated a classification report for each of the classifiers we ran our Bag of Words and RNN models against. Generally for the smaller data set, the RNN performed better than the classifiers. It would have performed better on the large top 3 diagnoses data as well if given more computation power and better fine tuning of the model. We also looked into the accuracy of the disease classification for different medical notes and looked to compare that accuracy with industry standards of predictive healthcare models. Additionally, we continued to create a well-developed and functional model that aligned with the project scope of implementing machine learning methodologies and data analysis to real-world EMR data.

4 Results

Our initial results for both Logistic Regression and Binary Naive Bayes reveal low macro accuracy of.60 and.53 respectively as shown in Fig. 1 and Fig. 2. The dummy classifier macro accuracy is.33 as shown in Fig. 3. Based on these results, the models are outperforming the baseline which indicates the model is learning, however, there is still low overall accuracy. This highlights the nuances and difficulties that can arise when working with complex and dense data to predict medical information. This result is significant in evaluating performance, so future iterations and approaches are strategically equipped for strong predictions.

Classification Report for Logistic Regression:

	precision	recall	f1-score	support
CONGESTIVE HEART FAILURE	0.63	0.58	0.61	2968
PNEUMONIA	0.56	0.58	0.57	3915
SEPSIS	0.59	0.60	0.60	3869
accuracy			0.59	10752
macro avg	0.60	0.59	0.59	10752
weighted avg	0.59	0.59	0.59	10752

Figure 1: This is the initial classification report for the Logistic Regression model.

With the refinement of our implementation, our results for the Logistic Regression model show a slight improvement of 0.03 in its f1-score as seen in Fig. 4 and nearly double in improvement for Binary Naive Bayes as seen in Fig. 5. This emphasizes the potential for the model to improve even further. In regards to the RNN, Fig. 6 shows the model training loss over 10 epochs. The top 3 diagnoses dataset was complex and therefore extremely difficult to correctly fine tune the hyper-parameters of the model to achieve high performing results. Our initial network achieved a .60 test accuracy however we wanted to fine tune the model to perform better. In trying to achieve that, our performance was not as good and reached a test accuracy of .5852. but we were able to better balance the distribution of the classes and more accurately represent

```
Classification Report for Binary Naive Bayes:
                          precision    recall  f1-score   support

CONGESTIVE HEART FAILURE       0.63      0.18      0.28      2968
               PNEUMONIA       0.58      0.13      0.21      3915
                  SEPSIS       0.39      0.90      0.54      3869

                accuracy                           0.42     10752
               macro avg       0.53      0.40      0.34     10752
            weighted avg       0.52      0.42      0.35     10752
```

Figure 2: This shows our initial classification report using Naive Bayes.

```
Classification Report for Dummy Baseline (Stratified):
                          precision    recall  f1-score   support

CONGESTIVE HEART FAILURE       0.29      0.29      0.29      2968
               PNEUMONIA       0.37      0.37      0.37      3915
                  SEPSIS       0.35      0.35      0.35      3869

                accuracy                           0.34     10752
               macro avg       0.33      0.33      0.33     10752
            weighted avg       0.34      0.34      0.34     10752
```

Figure 3: This shows our initial classification report using Dummy Classifier as the baseline model.

more parameters. Our neural network model still demonstrated improvement in learning as the loss graph showcases a decreased trend throughout the epochs. We tried extensive tuning with batch size and learning rate while adding gradient clipping for reduced overfitting. Overall with more epochs and greater computational power, we would be able to have a better-performing model to analyze from which would far surpass the regular logistic regression model.

```
Classification Report for Logistic Regression:
                          precision    recall  f1-score   support

CONGESTIVE HEART FAILURE       0.61      0.66      0.64      2968
               PNEUMONIA       0.60      0.59      0.59      3915
                  SEPSIS       0.63      0.61      0.62      3869

                accuracy                           0.61     10752
               macro avg       0.61      0.62      0.62     10752
            weighted avg       0.61      0.61      0.61     10752
```

Figure 4: This shows our final classification report for the Logistic Regression Model.

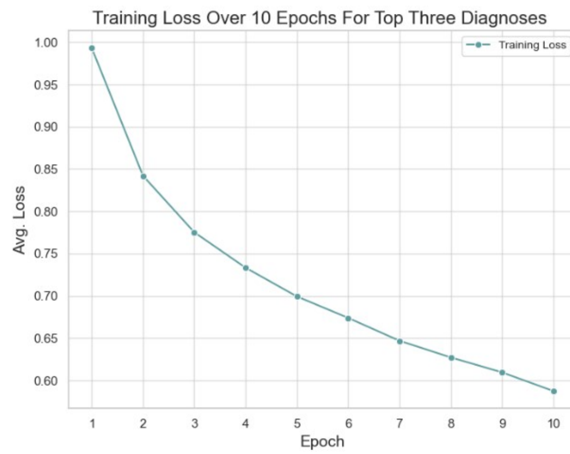
When applied to the data set filtered to three less common diagnoses (excluding extreme prematurity), the model showed tremendous progress in terms of performance. As shown in Fig. 5, while Binary Naive Bayes often performed far worse compared to Logistic Regression with fairly low scores across its classification report, it had moderate performance with a macro accuracy of 0.75 (previously 0.48). As for Logistic Regression, it achieved a moderately high macro accuracy of 0.83 and a f1-score of 0.81.

```
Classification Report for Binary Naive Bayes:
                          precision    recall  f1-score   support

CONGESTIVE HEART FAILURE       0.47      0.57      0.51      2968
               PNEUMONIA       0.53      0.10      0.16      3915
                  SEPSIS       0.44      0.73      0.55      3869

                accuracy                           0.45     10752
               macro avg       0.48      0.46      0.41     10752
            weighted avg       0.48      0.45      0.40     10752
```

Figure 5: This shows our final classification report using Naive Bayes.




```
Epoch 1/10, Avg, Loss: 0.9935, Accuracy: 0.5496
Epoch 2/10, Avg, Loss: 0.8422, Accuracy: 0.5750
Epoch 3/10, Avg, Loss: 0.7753, Accuracy: 0.5856
Epoch 4/10, Avg, Loss: 0.7334, Accuracy: 0.5774
Epoch 5/10, Avg, Loss: 0.6994, Accuracy: 0.5896
Epoch 6/10, Avg, Loss: 0.6740, Accuracy: 0.5899
Epoch 7/10, Avg, Loss: 0.6469, Accuracy: 0.5861
Epoch 8/10, Avg, Loss: 0.6272, Accuracy: 0.5860
Epoch 9/10, Avg, Loss: 0.6095, Accuracy: 0.5878
Epoch 10/10, Avg, Loss: 0.5875, Accuracy: 0.5844
Test Accuracy: 0.5852

Classification Report:
                          precision    recall  f1-score   support

CONGESTIVE HEART FAILURE       0.56      0.67      0.61      2968
               PNEUMONIA       0.58      0.53      0.56      3915
                  SEPSIS       0.62      0.58      0.60      3869

                accuracy                           0.59     10752
               macro avg       0.59      0.59      0.59     10752
            weighted avg       0.59      0.59      0.58     10752
```

Figure 6: This shows our Recurrent Neural Network training loss for the top three diagnoses.

In the training of the neural network for the less common diagnoses, the model performed substantially better as well. Fig. 9 showcases the training loss of the three less common diagnoses Renal Failure, Stroke, and Wound Infection over 20 epochs. This smaller dataset was able to perform faster and more efficiently enabling the training of more epochs and therefore more analysis into the performance of the model The loss decreases throughout each epoch and begins to level out in the later periods, indicating that the model has learned fairly well and reached a point of stability in its training. The model also effectively captured the patterns in this data with a .85 macro average and .8485 test accuracy which could be attributed to the more balanced target classes and better suited hyper-parameter tuning. Good prediction in less common diagnoses is also good for significantly improving earlier detection in serious conditions for patients.

Classification Report for Logistic Regression:

	precision	recall	f1-score	support
RENAL FAILURE	0.89	0.73	0.81	206
STROKE	0.81	0.87	0.84	188
WOUND INFECTION	0.78	0.87	0.82	187
accuracy			0.82	581
macro avg	0.83	0.82	0.82	581
weighted avg	0.83	0.82	0.82	581

Figure 7: This shows our classification report for the Logistic Regression model on the three less common diagnosis.

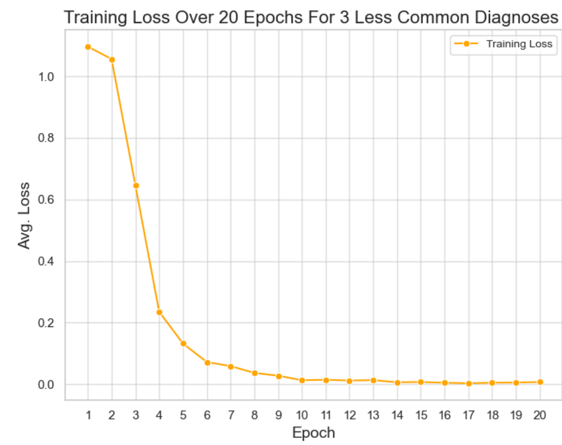
Classification Report for Binary Naive Bayes:

	precision	recall	f1-score	support
RENAL FAILURE	0.95	0.31	0.46	206
STROKE	0.65	0.91	0.75	188
WOUND INFECTION	0.65	0.87	0.75	187
accuracy			0.68	581
macro avg	0.75	0.70	0.65	581
weighted avg	0.76	0.68	0.65	581

Figure 8: This shows our classification report using Naive Bayes on the three less common diagnosis.

5 Discussion

The Neural Network model performed the best with an accuracy of .85 for the three less common diagnoses data. The final results of our Logistic Regression model with the application of Bag of Words, TF-IDF, and human rules showed .61 accuracy for the top three diagnoses and .82 for the three less common diagnoses. Though these results do not display high enough performance for the intended use of predicting diagnosis, the model still shows significance. The dummy model showed a performance accuracy of .34 which is essentially guessing out of the three diagnoses labels: Pneumonia, Sepsis, Congestive Heart Failure. Relative to the dummy model, the trained models show significant improvement. The model showed even greater results on the less common diagnosis labels: Renal Failure, Stroke, and Wound Infection. The models are nowhere near perfect or of quality for use in a practical setting, but they demonstrate the ability to improve performance through experimentation with NLP and machine learning algorithms.

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Epoch 1/20, Avg, Loss: 1.0967, Accuracy: 0.4172
Epoch 2/20, Avg, Loss: 1.0562, Accuracy: 0.7613
Epoch 3/20, Avg, Loss: 0.6463, Accuracy: 0.8258
Epoch 4/20, Avg, Loss: 0.2352, Accuracy: 0.8645
Epoch 5/20, Avg, Loss: 0.1321, Accuracy: 0.8559
Epoch 6/20, Avg, Loss: 0.0718, Accuracy: 0.8731
Epoch 7/20, Avg, Loss: 0.0586, Accuracy: 0.8796
Epoch 8/20, Avg, Loss: 0.0365, Accuracy: 0.8860
Epoch 9/20, Avg, Loss: 0.0274, Accuracy: 0.8731
Epoch 10/20, Avg, Loss: 0.0126, Accuracy: 0.8581
Epoch 11/20, Avg, Loss: 0.0141, Accuracy: 0.8710
Epoch 12/20, Avg, Loss: 0.0117, Accuracy: 0.8753
Epoch 13/20, Avg, Loss: 0.0130, Accuracy: 0.8624
Epoch 14/20, Avg, Loss: 0.0060, Accuracy: 0.8774
Epoch 15/20, Avg, Loss: 0.0069, Accuracy: 0.8602
Epoch 16/20, Avg, Loss: 0.0045, Accuracy: 0.8602
Epoch 17/20, Avg, Loss: 0.0028, Accuracy: 0.8645
Epoch 18/20, Avg, Loss: 0.0047, Accuracy: 0.8645
Epoch 19/20, Avg, Loss: 0.0053, Accuracy: 0.8452
Epoch 20/20, Avg, Loss: 0.0068, Accuracy: 0.8731
Test Accuracy: 0.8485
```

Classification Report:

	precision	recall	f1-score	support
RENAL FAILURE	0.88	0.80	0.84	206
STROKE	0.83	0.89	0.86	188
WOUND INFECTION	0.83	0.86	0.85	187
accuracy			0.85	581
macro avg	0.85	0.85	0.85	581
weighted avg	0.85	0.85	0.85	581

Figure 9: This shows our Recurrent Neural Network training loss for three less common diagnoses.

The main objective of this project was to focus on exploring different features to improve predictions in patients health. This was accomplished through the exploration of various strategies including Bag of Words, term frequency-inverse document frequency (TF-IDF), human rules, logistic regression, Naive Bayes, and neural networks. The implementation of these techniques showed improvements in our results indicating the ability to make better predictions in patients health with machine learning tools.

The results also highlight broader implications of what is possible with machine learning in the clinical setting. The results show significance that the models are able to learn from Medical Note data and make predictions based on the training. If this model could be built out more robustly, with more computing power, and more data, the results would likely increase as well. Our training only explored a subset of data and three potential diagnosis. There is much more data that exists regarding patient history and diagnosis that this model could learn from. The ability to submit a patients medical notes and confidently predict diagnosis with machine learning tools could combat the severe consequences of misdiagnosis that exist today. Additional benefits include potential acceleration of the diagnosis process and reduced waiting time for patients who are waiting for results. Letting technology identify your condition may seem untrustworthy, so a model like this can also act as an assistant, helping Doctors utilize hidden patterns in data and providing an additional opinion on challenging medical cases.

There is a relevant study that highlights the broader implications of using artificial intelligence(AI) in a clinical diagnostic setting. The study was performed at Beth Israel Deaconess Medical Center in Boston. Led by Dr. Adam Rodman the study split 50 doctors into two random groups. One group was instructed to use conventional resources (eg, UpToDate, Google) only, the other group was instructed to use the LLM interface (ChatGPT Plus [GPT-4]; OpenAI). Each participant was given six case histories and was asked to suggest diagnosis and provide explanation for their responses. The correctness of the final diagnosis was also included in the scoring [2].

The results were surprising as the capabilities of AI were expected to help doctors diagnose illness. The doctors who did not use ChatGPT scored 74% and those who used ChatGPT scored 76%. The use of ChatGPT had marginal impacts on the doctors performance. On top of this, ChatGPT was given the same cases and scored 15% higher with a score of 90%. These results can be attributed to the way doctors use AI. From evaluating the chat history the doctors using AI were not persuaded when the AI suggested a diagnosis that was different than their idea of the correct diagnosis [4]. Despite ChatGPT being a powerful resource, the doctors still chose to go with their individual reasoning. This study opens the door to many questions and considerations of AI in healthcare.

One consideration is the ability of AI to outperform professionals. Although this is a small study, the results display the capabilities of machine learning and AI very clearly. The study demonstrates that technology can be a great resource in the space of medical diagnosis. It could show greater impact if doctors know how to use AI to its fullest extent. This connects back to our project, aligning with the results that even simple and smaller scale machine learning models can demonstrate the ability to make predictions of diagnosis based on patterns in medical note data. There is reason to consider using this technology in the healthcare space as it is a powerful tool that can make a difference in patients health.

There are many ethical considerations as well. Since AI is a tool proven to make strong diagnostic predictions, what is the best and right way to integrate it into the current healthcare landscape. When should AI tools be trusted over professionals, if at all? How can it balance being a tool that aids professionals or one that professionals begin to rely on? Should patient data be used in these models or is it an obstruction of privacy and autonomy? These are just some of the many questions that surround this topic. Similar to the Rodman study, this project brings lights the the capabilities of AI and the existing problems it can combat, but also acknowledges the deep complexities when using AI in a sensitive setting like medicine.

There is strong potential for the effectiveness of AI in the healthcare landscape. By identifying methods and important features throughout the entire machine learning process, significant progress can be made. This technology could even expand to recommendation of prescription, procedures, and prognosis which have powerful implication in the field of medicine. The application of AI in healthcare diagnostics is a highly compelling area, as it necessitates addressing ethical considerations, the potential for human error, and the integration of transformative technologies. The ability to catch a disease early and eliminate misdiagnosis can save lives, time, and resources - improving the effectiveness and efficiency of the healthcare system as a whole.

6 Limitations

In this project, we encountered several limitations in development and performance. One major challenge was that the model, while capable of detecting patterns in the data, lacked the contextual understanding that healthcare professionals bring to decision-making. The model struggled to account for the nuanced nature of medical diagnoses, such as interpreting negative indicators, recognizing synonyms, or understanding the severity of symptoms. This highlighted the importance of integrating "human rules" into the model to improve its predictions and align its outputs more closely with clinical reasoning.

Another significant limitation was the computational demands of working with large-scale healthcare data. Filtering 2 million rows of electronic health records (EHR) to a manageable 54,000 rows required substantial processing power. Standard laptops struggled to handle the significantly reduced dataset efficiently, making it clear that more robust computational resources are necessary for scaling machine learning models in this domain. Especially properly training a neural network requires more computational power for effectively running numerous iterations of training. These challenges underscored both the technical and conceptual limitations we faced in developing an AI-driven diagnostic tool.

7 Conclusion

Our research highlights the potential of combining machine learning with natural language processing, NLP, to enhance the effectiveness and utility of electronic health records in predicting patient diagnoses. Through the implementation of Logistic Regression and Naive Bayes classifiers, along with Bag of Words (BoW) and Term Frequency-Inverse Document Frequency (TF-IDF) feature engineering, we achieved meaningful performance improvements in classification accuracy. Our models outperformed the baseline dummy classifiers demonstrating their ability to learn meaningful patterns in the data, especially in identifying patterns for the top three more frequent diagnoses, Congestive Heart Failure, Pneumonia, and Sepsis, and the 3 less common, Renal Failure, Stroke, and Wound Infection. Most notably our Logistic Regression model performed particularly well when applied to the less common diagnoses achieving a macro accuracy of .83 and an F1-score of .81. This improvement emphasizes the value of refining models with domain-specific "human rules" and customized stop-word dictionaries that are tailored to the complexities of medical notes. Moreover, the implementation of a Recurrent Neural Network (RNN) further elevated the predictive capabilities of our approach. By capturing the sequential nature of symptoms and medical notes, the RNN generally outperformed the traditional models specifically in complex cases where the contextual relationships between symptoms were more important. This model similarly performed the best with predicting less common diseases producing an overall accuracy of.85. This is extremely valuable for improving the early detection of serious illnesses and diseases.

However, like most models, there are several limitations to keep in mind. While the models performed generally well, the accuracy fell short of yielding consistent industry standards results of 85-95 percent accuracy range. Additionally, the challenging combination of dealing with high-dimensional medical data and the lack of computational processing power provided difficulties in fully contextualizing the performance of the models. Our results were based, in large, on the generalization of our models and therefore there remains work to be done to prevent further overfitting and deal with the underlying systemic, biases.

In a broader perspective, we demonstrated the potential for AI to potentially surpass and revolutionize current patient treatment care bu reducing human error and improving patient health. Despite this, it is important that these models are not only accurate but also exhibit the values of fairness, transparency and accountability in order to not exacerbate existing inequalities.

Overall, this research demonstrates the transformative potential of AI and machine learning in healthcare with the hopes in reducing diagnostic errors and significantly improving patient care. Specifically showcasing our progress in the iterative improvement of our models. However, careful attention to to ethical considerations and model limitations is necessary to fully understand how technology-driven data analysis ensures positive enhancements instead of hindering equity, and healthcare. With the immense power of innovative technologies comes the profound responsibility to ensure they serve the greater good in healthcare.

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